



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



OpenCTI curation: Normalize ASN labels

Threat Intelligence Report

Classification	TLP:CLEAR
Published	2026-06-16
Version	1.0
Author	Lost Boys Cyber
Report Type	Threat Intelligence Report

TLP:CLEAR | Lost Boys Cyber | OpenCTI curation: Normalize ASN labels - Threat Intelligence Report

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OpenCTI Curation Review: Normalize ASN Labels

1. Executive Summary

Lost Boys Cyber reviewed the OpenCTI hygiene candidate `opencti-curation-9021605476763b87`, which flagged Autonomous System Number (ASN) strings being used as taxonomy labels. The audit sample identified 12 ASN-like labels in a 500-label sample; the current portal task scoped 10 specific OpenCTI label objects for review. The evidence supports the candidate finding: these labels describe infrastructure context attached to observables and indicators, not durable analyst taxonomy.

The reviewed labels should not be deleted blindly. They currently preserve useful context for botnet command-and-control, RAT, Cobalt Strike, Mythic, Sliver, Havoc, and related infrastructure records. Lost Boys Cyber should normalize them by migrating ASN context into infrastructure, identity, location, relationship, or observable-enrichment fields while retaining analyst-facing tags such as malware family, C2 framework, source, and infrastructure role.

Field	Assessment
Report family	Threat Intelligence Report
Curation candidate	`opencti-curation-9021605476763b87`
Category	Label normalization
Severity	Medium
Proposed action	Manual review; safe-mode normalization only
Source artifact	`/home/lost0x01/services/opencti-deploy/audit/latest-opencti-hygiene-audit.json`
Audit generated	2026-06-16T03:21:38Z
Analyst judgment	ASN labels are useful enrichment but weak taxonomy; migrate context before retiring labels
Confidence	Moderate-High for data-model issue; Moderate for downstream object coverage due to sampled audit scope

2. Intelligence Requirement

This review answers the following internal source-of-truth questions:

Requirement	Answer
Are ASN strings being used as OpenCTI taxonomy labels?	Yes. The hygiene audit found 12 ASN-like labels in a 500-label sample, and the portal task scoped 10 for review.
Are the labels attached to relevant threat-intelligence objects?	Yes. The reviewed labels are attached to Indicators, IPv4 observables, and at least one Malware object.
Should these labels remain as durable taxonomy?	No. ASN values are infrastructure attributes and should be modeled as infrastructure/identity/location context or observable enrichment, not as permanent taxonomy labels.
Can the labels be removed immediately?	No. They currently carry useful context and must be migrated or replaced before deletion.
What is the safe next action?	Create a reviewed normalization backlog, map each ASN label to

	canonical infrastructure context, update affected objects, then remove or deprecate only verified redundant labels.
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3. Source Review and Confidence

Primary evidence came from the local OpenCTI hygiene audit artifact and authenticated OpenCTI GraphQL checks against the scoped label objects. The audit reported a healthy OpenCTI platform and strong overall data volume, including 45,767 indicators, 44,204 STIX cyber observables, 4,356 labels, 768 reports, 4,433 malware objects, 442 intrusion sets, and 1,735 vulnerabilities. This means the curation issue is not a platform availability issue; it is a taxonomy and data-model quality issue inside an otherwise populated TIP.

Source	Evidence used	Reliability	Limitations
`latest-opencti-hygiene-audit.json`	Counts, sampled label-noise finding, ASN candidate IDs and values	High for sampled audit output	Sampled labels only; not a full taxonomy census
OpenCTI GraphQL checks	Confirmed object usage counts and sample attached objects for scoped labels	High for live local OpenCTI state at review time	Query results can change as connectors import new data
Operator guidance	Stated model preference: represent ASNs through infrastructure/identity/location context rather than taxonomy	High for house data-model intent	Requires implementation decision before mutation

Confidence is Moderate-High because the sampled labels match a clear pattern (`as` followed by digits) and were observed on infrastructure-adjacent objects. The recommendation remains conservative because the labels still support investigation and should be migrated, not discarded.

4. Key Findings

Finding	Evidence	Impact	Priority
ASN labels are being used as attribute labels	Audit found 12 ASN-like labels in a 500-label sample; 10 scoped IDs were reviewed	Blurs the distinction between taxonomy and infrastructure enrichment	Medium
ASN labels are attached to operationally relevant objects	Live checks showed the labels attached to Indicators, IPv4 observables, and one Malware object	Blind deletion would remove triage context from C2 and malware infrastructure records	Medium
Label colors appear arbitrary	Each ASN label carries a separate color value with no analytic semantics	Visual noise in OpenCTI and inconsistent analyst experience	Low-Medium
Labels duplicate better modeling options	ASNs can be represented through Infrastructure, Autonomous System / identity context, location, external references, or observable enrichment	Improves graph accuracy and query consistency	Medium
The issue is part of a broader label-noise pattern	Same audit sample also found 30 dashed-IP labels and 0 CVE labels	Indicates imported feed metadata may be promoted to labels too aggressively	Medium

Scoped ASN Labels Reviewed

Label value	OpenCTI label ID	Live usage count	Sample object context	Analyst disposition
`as8560`	`2444e6aa-7836-476b-8449-8d67bd244ff3`	2	Indicator/IPv4 `85.215.57.133`; labels included `adaptixc2`, `botnet_cc`, `c2`, `censys`, `ionos-as`	Migrate to infrastructure/ASN context; retain C2/tool/source labels
`as5400`	`a5a8f3a3-3ff0-488d-bd6d-a46d567f590b`	2	Indicator/IPv4 `195.100.198.220`; labels included `botnet_cc`, `bt`, `c2`, `censys`, `mythic`	Migrate ASN context; preserve Mythic/C2/source tags
`as13124`	`792ac248-e435-4b30-9933-ce7f6c41cd3d`	2	Indicator/IPv4 `85.130.116.122`; labels included `a1bg_rsd`, `botnet_cc`, `censys`, `chaos`, `panel`	Migrate ASN context; preserve panel/tool/source tags
`as48282`	`2505f15a-2dc1-4cf4-b76d-695c6fc35c87`	5	Indicators/IPv4 `88.218.60.191`, `178.217.98.23`; labels included `adaptixc2`, `botnet_cc`, `c2`, `vdsina-as`	Prioritize due to multiple affected records
`as12876`	`4ecee750-614f-4a77-a7e6-820c8692808b`	2	Indicator/IPv4 `51.15.8.6`; labels included `botnet_cc`, `c2`, `online`, `sliver`	Migrate ASN context; preserve Sliver/C2/source tags
`as62563`	`ac9da208-ba97-4453-8c0b-73dabcb96e1b`	4	Indicators/IPv4 `5.101.86.72`, `5.101.86.44`; labels included `as-globaltelehost`, `botnet_cc`, `c2`, `rat`	Migrate numeric ASN; review whether provider label should become identity/infrastructure context
`as134768`	`cf055080-4fcc-425b-a7d2-242c3ea23a2a`	3	Indicator/IPv4 `103.236.70.158` and Malware object; labels included `chinanet-shaanxi-cloud-base`, `dcrat`, `botnet_cc`	Normalize carefully because a Malware object also carries the label
`as9304`	`ddfd02f3-ff96-4fca-99c4-573c9f6db5de`	2	Indicator/IPv4 `196.251.107.104`; labels included `asyncrat`, `botnet_cc`, `c2`, `rat`, `triage`	Migrate ASN context; preserve AsyncRAT/RAT/triage tags
`as45090`	`5457f906-2135-41d7-9d56-ffbb0ba8a4ad`	30	Multiple Cobalt Strike indicators including `118.25.85.198`, `118.89.88.183`,	Highest priority due to usage volume and Cobalt Strike context

			`150.158.170.241`	
`as211138`	`135cceb5-35f8-42fe-82e5-b6464769f8e4`	2	Indicator/IPv4 `41.216.189.77`; labels included `havoc`, `privatehosting-net`, `botnet_cc`, `c2`	Migrate ASN context; preserve Havoc/C2/source tags

5. Threat Context

The affected labels are not random metadata. They appear on infrastructure records associated with C2 panels, RATs, red-team frameworks abused by threat actors, and botnet command-and-control tracking. The labels therefore have defensive value, but their current representation creates long-term source-of-truth problems.

ASN context is volatile and belongs close to the infrastructure or observable. If an IP address moves, a provider changes ownership, or a feed corrects enrichment, a label such as `as45090` does not explain when the association was observed, which source asserted it, or whether it applies to all linked objects. Treating ASNs as labels also makes it harder to distinguish analytic categories (`botnet\_cc`, `c2`, `rat`, `cobaltstrike`) from enrichment fields (`as45090`, provider strings, geographic strings, scanned-by-source labels).

Threat-intel use case	Why ASN context matters	Why label modeling is weak
C2 infrastructure clustering	Helps identify hosting concentration and repeat provider abuse	Labels do not capture observation time, source, confidence, or relationship direction
Provider-risk triage	Helps analysts identify high-noise hosting or bulletproof provider patterns	Provider/ASN strings become mixed with malware/tool labels
Detection enrichment	SOC queries may group alerts by ASN or hosting provider	SIEM fields should use normalized ASN/provider enrichment, not ad hoc OpenCTI label text
Graph exploration	Infrastructure context can link indicators, observables, and reports	Labels create flat tag clouds instead of typed relationships

MITRE ATT&CK mapping is indirect because this is a curation finding, not a specific intrusion. The affected objects support analysis of adversary infrastructure and C2 activity, which commonly maps to:

ATT&CK technique	Relevance	Confidence
T1583 - Acquire Infrastructure	ASN/provider context can support infrastructure acquisition and hosting analysis	Medium
T1584 - Compromise Infrastructure	Infrastructure clustering may reveal abused or compromised hosting	Low-Medium without case-specific evidence
T1071 - Application Layer Protocol	Many C2 records ultimately relate to network C2 behavior	Low-Medium; protocol evidence not reviewed in this curation task
T1090 - Proxy	ASN/provider context can assist proxy/VPS triage	Low unless object-specific evidence supports proxy behavior

6. Detection and Hunting

This curation task has two detection dimensions: detecting taxonomy drift inside OpenCTI and using normalized ASN context for SOC hunting after migration.

6.1 OpenCTI Taxonomy Drift Detection

Run this style of query before any label deletion to enumerate affected objects and confirm migration candidates.

```
query ObjectsByAsnLabel($filters: FilterGroup!) {
  stixCoreObjects(first: 100, filters: $filters) {
    pageInfo { globalCount }
    edges {
      node {
        id
        entity_type
        created_at
        updated_at
        objectLabel { id value }
        ... on Indicator { name pattern x_opencti_score confidence }
        ... on StixCyberObservable { observable_value }
      }
    }
  }
}
```

Example filter body:

```
{
  "filters": {
    "mode": "and",
    "filters": [
      {
        "key": "objectLabel",
        "values": ["5457f906-2135-41d7-9d56-ffbb0ba8a4ad"],
        "operator": "eq",
        "mode": "or"
      }
    ],
    "filterGroups": []
  }
}
```

Use this audit check to find newly imported ASN-like labels after connector runs:

```
import re
asn_label = re.compile(r"^as\d{1,10}$", re.IGNORECASE)
noisy = [label for label in labels if asn_label.match(label["value"])]
for label in noisy:
  print(label["id"], label["value"], label.get("color"))
```

6.2 SOC Hunting After ASN Normalization

If endpoint or network telemetry enriches remote connections with ASN fields, hunt on normalized ASN numbers and then pivot to known OpenCTI C2/tool context. The following examples should be adapted to the field names used by the deployed SIEM.

```
let ReviewedASNs = dynamic([8560,5400,13124,48282,12876,62563,134768,9304,45090,211138]);
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where RemoteIPType == "Public"
| where RemoteIPASN in (ReviewedASNs)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Connections=count(),
Devices=dcount(DeviceName), Users=dcount(InitiatingProcessAccountName) by RemoteIPASN, RemoteIP,
RemoteUrl, InitiatingProcessFileName
| order by Connections desc

index=network earliest=-30d
| where isnotnull(dest_asn)
| search dest_asn IN (8560,5400,13124,48282,12876,62563,134768,9304,45090,211138)
| stats min(_time) as first_seen max(_time) as last_seen count dc(src_host) as src_hosts values(app) as
apps by dest_asn dest_ip dest_host
| convert ctime(first_seen) ctime(last_seen)
| sort - count
```

```

logsource:
  product: network
  category: network_connection
detection:
  selection:
    DestinationASN:
      - 8560
      - 5400
      - 13124
      - 48282
      - 12876
      - 62563
      - 134768
      - 9304
      - 45090
      - 211138
    condition: selection
  fields:
    - SourceHostname
    - SourceIp
    - DestinationIp
    - DestinationHostname
    - DestinationASN
    - ProcessName
falsepositives:
  - Legitimate business traffic to cloud, hosting, CDN, or VPS providers using the same ASNs
level: low

```

6.3 Analyst Validation Checklist

Check	Purpose	Expected outcome
Enumerate objects by label ID	Prevent accidental context loss	Every affected object is known before mutation
Compare labels to object type	Identify labels on Malware/Report objects that need special handling	Numeric ASNs are removed only where replacement context exists
Validate indicator patterns	Ensure the affected records are IP/network indicators, not unrelated entities	ASN context is migrated to observables/infrastructure
Preserve tool/family labels	Retain analyst taxonomy such as `cobaltstrike`, `sliver`, `mythic`, `asynrat`, `havoc`	Taxonomy remains useful after ASN cleanup
Re-run hygiene audit	Confirm normalization reduced ASN labels without damaging source data	`asn_labels` count decreases or candidates are marked reviewed/skipped

7. Recommendations

Priority	Recommendation	Owner	Acceptance criteria
P1	Treat the 10 scoped labels as manual-review normalization candidates, not deletion candidates	TIP curator	No label is deleted until affected objects are enumerated and replacement context is recorded
P1	Normalize `as45090` first because it had the highest observed usage count and Cobalt Strike context	TIP curator / detection engineer	All 30 affected objects have reviewed ASN/provider context and retained C2/tool labels

P2	Add an OpenCTI hygiene rule that flags labels matching <code>`^as\d+\$`</code> after connector imports	Platform operator	Weekly audit outputs stable candidate IDs and safe-mode review tasks
P2	Model ASNs as infrastructure/identity/location or observable enrichment, with source and observation time where available	TIP curator	Analysts can query affected infrastructure without relying on flat labels
P2	Preserve meaningful analyst taxonomy labels while removing atomic enrichment labels	Threat intel analyst	Malware/tool/source labels remain attached after migration
P3	Review dashed-IP labels from the same audit as a separate curation queue	TIP curator	IP-like labels are converted to IPv4 observables or marked as source-specific evidence
P3	Document the house OpenCTI modeling rule for ASN, IP, CVE, provider, and geolocation metadata	Platform owner	Future imports do not promote atomic observables into durable labels by default

Suggested implementation sequence:

- Export the affected object list for each label ID and store it as curation evidence.
- For each affected IP indicator/observable, verify whether OpenCTI already has corresponding infrastructure, location, or enrichment fields.
- Create or link typed infrastructure context where it is absent.
- Remove the numeric ASN label only after the replacement context is visible and queryable.
- Re-run ``opencti_hygiene_audit.py`` and compare ``label_noise.asn_labels`` and object counts.
- Keep a reviewed skip list for cases where the label must remain temporarily because OpenCTI lacks a suitable replacement relationship.

Indicators of Compromise

No source-validated malicious atomic indicators were disclosed for this work item. Lost Boys Cyber therefore treats this section as an IOC disposition record so downstream publication, OpenCTI hygiene, and detection-engineering workflows do not create placeholder blocklist entries.

Indicator Type	Value	Disposition
OpenCTI object / label IDs	See scoped object IDs and label table above	Normalize as CTI data-model context; do not treat taxonomy labels as network/blocklist IOCs.
Workflow context	threat-intel / opencti-curation	Use the report's detection and hunting logic for monitoring; do not promote non-malicious metadata into IOC-only controls.

8. References

- [1] Lost Boys Cyber OpenCTI hygiene audit artifact: ``/home/lost0x01/services/opencti-deploy/audit/latest-opencti-hygiene-audit.json``, generated 2026-06-16T03:21:38Z.

- [2] OpenCTI live GraphQL review against scoped label IDs from candidate `openciti-curation-9021605476763b87`, performed 2026-06-16 UTC.
- [3] STIX 2.1 data-model principle: represent cyber observables, infrastructure, identities, locations, and relationships as typed objects rather than flat taxonomy where practical.
- [4] MITRE ATT&CK Enterprise techniques T1583, T1584, T1071, and T1090 for infrastructure and C2 analytic context; mappings in this report are contextual rather than incident-specific.